

This assignment will not be graded and is only for practice.

Let $T(x, y, z) = mx + ny + pz$ denote the total amount of money a multiplex made from a specific movie, in one day by selling x number of tickets for the morning show, y number of tickets for the evening show, and z number of tickets for the night show, where m , n and p are the prices of each ticket of that movie in the morning, evening, and night show, respectively.

In the multiplex, 2 Malayalam movies, 1 Hindi movie, 1 Bengali movie, and 1 English movie are running parallelly in one week. Suppose there are 3 shows (Morning, Evening, Night) per day for each of the Malayalam movies and the English movie, whereas 2 shows (Evening, Night) per day for the Hindi and the Bengali movie. Ticket price (irrespective of morning, evening, and night shows) for each Malayalam movie is ₹200, Hindi movie is ₹250, Bengali movie is ₹250, and English movie is ₹300. The number of tickets sold in a particular day is given in Table M2W5AQ2:

Shows	Morning	Evening	Night
Malayalam Movie 1	m_1	m_2	m_3
Malayalam Movie 2	m'_1	m'_2	m'_3
Hindi Movie	0	h_2	h_3
Bengali Movie	0	b_2	b_3
English Movie	e_1	e_2	e_3

Table: M2W5AQ2

Answer questions 1 to 8 using the given data above.

Level 1

1) Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the morning shows only? **1 point**

- ☐ $200m_1 + 200m_2 + 200m_3$
- ☐ $200m_1 + 250m'_1 + 300e_1$
- ☐ $200m_1 + 200m'_1 + 250e_1$
- ☐ $200m_1 + 200m'_1 + 300e_1$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$200m_1 + 200m'_1 + 300e_1$$

2) Which expression accurately expresses the total amount of money the multiplex made in the **1 point** specific day mentioned above by selling the tickets of the Bengali movie only?

- ☐ $200(b_2 + b_3)$
- ☐ $250(b_2 + b_3)$
- ☐ $250(h_2 + h_3)$
- ☐ $250b_2 + 200b_3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$250(b_2 + b_3)$$

3) Which expression accurately expresses the total amount of money the multiplex made in the **1 point** specific day mentioned above by selling the tickets of the night shows only?

- ☐ $200(m_2 + m'_2) + 250(h_2 + b_2) + 300e_2$
- ☐ $200(m_3 + m'_3 + h_3 + b_3) + 300e_3$
- ☐ $200(m_3 + m'_3) + 250(h_3 + b_3) + 300e_3$
- ☐ $200(m_3m'_3) + 250(h_3b_3) + 300e_3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$200(m_3 + m'_3) + 250(h_3 + b_3) + 300e_3$$

4) Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Bengali and English movie only? **1 point**

- ☐ $250(b_2 + b_3)$
- ☐ $300e_1 + 250(b_2 + e_2) + 300(b_3 + e_3)$
- ☐ $250(b_2 + b_3) + 300(e_1 + e_2 + e_3)$
- ☐ $b_2 + b_3 + e_1 + e_2 + e_3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$250(b_2 + b_3) + 300(e_1 + e_2 + e_3)$$

Level 2

5) What was the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Malayalam movies together? **1 point**

- ☐ $T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3)$
- ☐ $200T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3)$
- ☐ $T(m_1m'_1, m_2m'_2, m_3m'_3)$
- ☐ $200T(m_1m'_1, m_2m'_2, m_3m'_3)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3)$$

6) Which of the following options are correct?

1 point

- ☐ $T(x, y, z) = T(x, 0, 0) + T(0, y, 0) + T(0, 0, z)$
- ☐ $T(x, y, z) = T(x, y, 0) + T(0, 0, z)$
- ☐ $T(x, y, z) = T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0)$
- ☐ $T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0) = T(x + y + z, 0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(x, y, z) = T(x, 0, 0) + T(0, y, 0) + T(0, 0, z)$$

$$T(x, y, z) = T(x, y, 0) + T(0, 0, z)$$

$$T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0) = T(x + y + z, 0, 0)$$

7) Which of the following equalities correctly represent the total amount of money the multiplex **1 point** made in the specific day mentioned above by selling the tickets of the Bengali and Hindi movie together?

- ☐ $T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), (h_3 + b_3))$
- ☐ $T(0, b_2 + h_2, b_3 + h_3) = T(0, 0, 0) + T(0, h_2 + b_2, 0) + T(0, h_3 + b_3, 0)$
- ☐ $T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), 0) + T(0, 0, (h_3 + b_3))$
- ☐ $T(0, b_2 + h_2, b_3 + h_3) = T(0, b_2, b_3) + T(0, h_2, h_3)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), (h_3 + b_3))$$

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, 0, 0) + T(0, h_2 + b_2, 0) + T(0, h_3 + b_3, 0)$$

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), 0) + T(0, 0, (h_3 + b_3))$$

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, b_2, b_3) + T(0, h_2, h_3)$$

8) Which of the following equality correctly represents the total amount of money the multiplex **1 point** made in the specific day mentioned above by selling the tickets of the English movie?

- ☐ $T(e_1, e_2, e_3) = 200(e_1 + e_2 + e_3)$

☐ $T(e_1, e_2, e_3) = 250(e_1 + e_2 + e_3)$

☐ $T(e_1, e_2, e_3) = 200e_1 + 250e_2 + 300e_3$

☐ $T(e_1, e_2, e_3) = 300(e_1 + e_2 + e_3)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(e_1, e_2, e_3) = 300(e_1 + e_2 + e_3)$

Your score is: 0/8